AGRICULTURAL DRAINAGE · NOTE DR07

Mole, Vertical and Bio-Drainage — Mole Channels, Drainage Wells and Reclaiming Salt-Affected Soils

Course topics: IDE-502 T6, IDE-502 T9, IDE-502 T13, IDE-502 T14, IDE-511 T4, MDC-IV L19

Mole drainage

- Mole drains are unlined circular or oval earthen channels formed underground in highly cohesive or fibrous soils by a **mole plough** — a long blade-like shank carrying a cylindrical, bullet-nosed plug (the mole), drawn by a high-powered prime mover.
- The plough leaves a cavity at the set depth and fractures the soil, giving water escape routes. The channel receives, conducts and discharges water to field drainage ditches.
- **Not effective** in very loose soils (the channel collapses) or heavy plastic soils (sealing stops water movement).

Item	Value
Diameter	7.5–10 cm
Spacing	1.5–9 m
Depth	50–75 cm; usually formed 40–60 cm below the surface, within the impermeable subsoil
Soil	Clay soils with > 30% clay
Safe length	20–80 m, depending on soil
Safe gradient	0.2%–3.0%
Re-laying interval	Every 3–7 years
Outlet protection	1–2 m long pipe inserted in the mole channel

- Lay out like tile drains, usually aligned with the land slope; spacing and depth are generally less than for tiles. Outlets go into open ditches, tile lines or other mole channels; metal tubing can protect an outlet into an open ditch.
- **Best time:** soon after harvest, when the upper profile has dried.
- **Life:** depends on the soil's stability under repeated wetting, its uniformity and conditions at installation. Longer moles block more easily; stagnant water weakens the walls. Well-installed drains work for 3–5 years, then less well for another 3–5 years; cultivation shortens their life, and they can be redrawn to restore efficiency. Durability depends on soil type, outlet adequacy, length, slope, depth, spacing and moisture at construction.
- **Slope:** a good outlet and some slope are needed, but too much slope erodes the channel.
- **Collectors:** moles may discharge straight into an open channel (protected or not). Better: lay pipe drains in trenches across the mole lines first, then draw the moles, and backfill the trenches above mole depth with clean 3–5 mm gravel. Moles respond quickly to rain and can discharge heavily, so size collector pipes accordingly.

- **Aim:** a medium-deep drainage base in the soil. Widely practised in England, Western Europe and New Zealand.

Soil failure and critical depth

- At shallow depth the soil is pushed forward, sideways and upward and shears along rupture planes (~45°) from the mole top to the surface — the cracking raises porosity and hydraulic conductivity.
- Below the **critical depth** the soil at the base moves sideways and is compacted, so porosity and conductivity fall.
- Place the mole just below the critical depth, so the channel lies below it and a connected zone of cracked soil reaches the surface to let water in.
- The notes give the critical depth as an aspect ratio (depth : thickness) of "≈ 5:7".

Check: tine studies (Spoor and Godwin, 1978) put the critical depth at about 5–7 times the tine width, so "5:7" is best read as a depth-to-width ratio of 5 to 7. The ICAR eCourse adds: normal mole diameter 5–10 cm (about 10 cm usual), moling when soil moisture lies between the liquid and plastic limits, and a maximum life of 10–30 years in favourable soils (Schwab et al., 2005).

Vertical drainage

- Control the water table by pumping from a network of wells. Pumping must be from shallow depths hydraulically connected to the root-zone water; shallow pumping has a small cone of depression, so many pumping points are needed.
- In deep alluvial aquifers with good-quality water and a rising water table, deep wells are used; the pumped water can augment the canal supply or irrigate directly, and preventing a high water table gives an indirect drainage effect.

Bio-drainage

Plants with high groundwater use — e.g. eucalyptus and poplar — draw down the water table. Their water use is limited, so bio-drainage suits physically and chemically degraded land as an alternative use rather than full reclamation. Vertical drainage and bio-drainage are relatively new terms.

Drainage wells

- Widely used to drain farmland, especially irrigated areas. Water-table (gravity) wells in an unconfined aquifer remove root-zone water; deep tubewells tap confined aquifers.
- Design follows well-design principles (Chapter 6). The more adjacent wells interfere, the greater the drainage effect. The permeability of the layers to be drained is critical: drainage water must percolate to the pumped aquifer, the water table must be continuous from the shallow layers to the aquifer, and no impermeable barrier may intervene. Most successful in deep, highly permeable sandy soils.
- **Site investigation:** study the soil profile and the aquifer's thickness and extent with test holes and pumping tests; the aquifer must be extensive enough to allow large withdrawals.
- **Other uses:** water-table control, extra irrigation supply, and pumping canal seepage or water from waterlogged zones back into the canal. Assess economic feasibility and the long-term water-table impact before large projects.

Draining salt-affected soils

Soil	Treatment
Saline	Common in arid and semi-arid regions; usually improved by **leaching** — soluble salts dissolve and leave with the drainage water. In high-rainfall areas leaching may happen naturally once surface drainage is in place; in arid and semi-arid regions irrigation water must be supplied for leaching
Non-saline alkali	Strongly alkaline soils become nearly impermeable, so apply **chemical amendments before** leaching and drainage — calcium chloride, gypsum (calcium sulphate), sulphur or sulphuric acid, the type and amount set by laboratory soil analysis. Because reclamation is complex and costly, it may not always pay
Saline-alkali	Treat like non-saline alkali soils, with chemicals chosen from laboratory analysis

Sources: ICAR eCourse 'Drainage Engineering' — Dr M. K. Jha (IIT Kharagpur), reviewed by Prof. K. Yellareddy (Director A&R, WALAMTARI, Hyderabad); AgriMoon e-book. Spoor, G. and Godwin, R. J. (1978), An experimental investigation into the deep loosening of soil by rigid tines, J. Agric. Eng. Res. 23: 243–258. Further reading: A. M. Michael and T. P. Ojha, Principles of Agricultural Engineering, Vol. II (Jain Brothers).

Practice MCQs

Q1. Mole drains are best suited to soils with clay content above

(a) 5%
(b) 15%
(c) 30%
(d) 60%

Q2. Typical mole drain spacing is

(a) 0.5–1 m
(b) 1.5–9 m
(c) 15–30 m
(d) 30–60 m

Q3. Safe gradient for mole drains is

(a) 0.01–0.05%
(b) 0.2–3.0%
(c) 5–8%
(d) 10–15%

Q4. Moles are usually re-laid every

(a) 1 year
(b) 3–7 years
(c) 15–20 years
(d) 50 years

Q5. Below the critical depth, a mole plough

(a) cracks the soil upward
(b) compacts the soil sideways
(c) has no effect
(d) creates surface ridges

Q6. Vertical drainage controls the water table by

(a) open ditches
(b) pumping from wells
(c) trees
(d) mole channels

Q7. Before leaching a non-saline alkali soil you must

(a) add gypsum or another amendment
(b) add nitrogen
(c) irrigate with saline water
(d) plant eucalyptus

Q8. Drainage wells work best in

(a) heavy clay with a hard pan
(b) deep, highly permeable sandy soils
(c) peat over rock
(d) shallow saline crusts

Answers

Q1 — (c) 30%
Q2 — (b) 1.5–9 m
Q3 — (b) 0.2–3.0%
Q4 — (b) 3–7 years
Q5 — (b) compacts the soil sideways
Q6 — (b) pumping from wells
Q7 — (a) add gypsum or another amendment
Q8 — (b) deep, highly permeable sandy soils
